HDOC 14-1

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Program to Print the Product of Even Numbers from 1 to n

This document provides a clean, well-commented implementation for calculating and printing the product of even numbers from 1 to **n** in C. The program takes the value of **n** as input and uses a for loop to compute the cumulative product.

1. C Implementation

Step 1: Understand the Logic

A for loop repeats a block of code while its condition is true. To find the product of even numbers from 1 to **n**, the loop iterates through numbers from 1 up to **n**, checks if a number is even, and multiplies it into a running product.

- **Initialization:** `product = 1` (since multiplying by 0 would make the result 0) and i = 1, so counting starts from 1.
- **Condition:** i ≤ n, so the loop keeps running until i becomes greater than **n**.
- **Update:** `i++`, so i increases by 1 after every pass through the loop.
- Inside the loop, if i % 2 == 0, the number is even and is multiplied by `product`.
- If **n** is less than 2, there are no even numbers to multiply, so the input is rejected with an error message.

Step 2: Define Variables and Data Types

Use `int` for the limit (n) and loop counter (`i`), and `long long` for the accumulated product (`product`).

Step 3: Take User Input

Display a prompt using `printf("Enter the value of n: ");`, then read the value using `scanf("%d", &n);`.

Step 4: Validate the Input

If n < 2, print an error message and do not run the loop.

Step 5: Calculate the Product Using a for Loop

1. Start with i = 1 and `product = 1`.
2. While i ≤ n, check if i is even. If yes, multiply it to `product`.
3. Increase i by 1 and check the condition again.
4. When i becomes greater than **n**, the loop ends. Print the final product.

Complete C Code

```
/* * Q23 (User Inputs, Operations & Output) * Write a program to print the product of even numbers from 1 to n. */ #include <stdio.h> int main() { int n, i; long long product = 1; // Step 1: Read the value of n printf("Enter the value of n: "); scanf("%d", &n); // Step 2: Validate n if (n < 2) { printf("Invalid input! n must be at least 2 to have even numbers.\n"); } else { // Step 3: Calculate product of even numbers using a for loop for (i = 1; i <= n; i++) { if (i % 2 == 0) { product *= i; } } printf("The product of even numbers from 1 to %d is: %lld\n", n, product); } return 0; }
```

Program Output (Sample Runs)

```
Run 1: $ gcc hdoc14_1.c -o hdoc14_1 $ ./hdoc14_1 Enter the value of n: 5 The product of even numbers from 1 to 5 is: 8 Run 2: $ ./hdoc14_1 Enter the value of n: 6 The product of even numbers from 1 to 6 is: 48 Run 3: $ ./hdoc14_1 Enter the value of n: 1 Invalid input! n must be at least 2 to have even numbers.
```

Dry Run with Test Cases

Sample Test Case 1:
Input: n = 5
Calculation:

- 5 < 2 is FALSE, so the input is valid and the loop runs.
- Start: i = 1, `product = 1`.
- Pass 1: 1 ≤ 5 is TRUE, 1 is not even, i becomes 2.
- Pass 2: 2 ≤ 5 is TRUE, 2 is even, `product = 1 × 2 = 2`, i becomes 3.
- Pass 3: 3 ≤ 5 is TRUE, 3 is not even, i becomes 4.
- Pass 4: 4 ≤ 5 is TRUE, 4 is even, `product = 2 × 4 = 8`, i becomes 5.
- Pass 5: 5 ≤ 5 is TRUE, 5 is not even, i becomes 6.
- Check: 6 ≤ 5 is FALSE, loop ends.

Output: The product of even numbers from 1 to 5 is: 8

Sample Test Case 2:
Input: n = 1
Calculation: 1 < 2 is TRUE, so the input is invalid. Error message is printed and loop does not run.
Output: Invalid input! n must be at least 2 to have even numbers.